

Test specification — Switch It

QA / test plan for the course assignment.
It describes what should be tested, what the repo already covers automatically, and what you should still verify manually.

Test runner (actual): Vitest + React Testing Library + jsdom.
Not a project test runner in package.json : Playwright, Cypress, Jest.

Re-verified during this documentation pass:

```
npx vitest run
→ 277 test files passed
→ 1744 tests passed

npx eslint .
→ exit 0

npm run build
→ Next.js 16.2.12 compiled successfully
```

Coverage labels:

- **Automated** — evidence in *.test.ts(x) / migration contract tests
- **Manual** — run with two accounts before submission
- **Missing / weak** — little or no automated proof

Important honesty: Migration tests that inspect SQL text are **not** the same as running true concurrent transactions against live PostgreSQL. Concurrent claim safety is **implemented in the database**; live multi-client verification is still useful.

Testing strategy

Layer	Tool	What it covers
Unit / component	Vitest + Testing Library	UI components, helpers, action wrappers (often mocked Supabase)
Migration contract	Vitest reading SQL files	Asserts important SQL phrases exist in migrations
Live DB integration / true parallel concurrency	Not a first-class automated project suite	Use manual two-user races
Full browser E2E automation	Not currently implemented	Manual checklist

Authentication tests

Case	Expected	Coverage
Register / login validation	Invalid input rejected	Automated
Password policy on register	Under 8 / missing upper/lower/digit/special rejected	Automated
Signup with Confirm email	No session → Check your email (neutral inbox copy + Sign in hint)	Automated
Obfuscated existing-email signup	Neutral Check your email; no false “account created” certainty	Automated
Explicit duplicate Auth error	“An account with this email already exists...”	Automated
Resend verification	Neutral delivery wording; unconfirmed users can still resend	Automated
Unconfirmed login	Friendly verify/resend state	Automated
Forgot password link on login	Navigates to /forgot-password	Automated
Forgot-password request	resetPasswordForEmail with redirectTo → /auth/callback/recovery; neutral Check your email; no email enumeration	Automated
Reset rate limit	Friendly “Too many attempts...”	Automated
Recovery callback	Reset email → /auth/callback/recovery?code=... → exchangeCodeForSession → /auth/reset-password (dedicated route; Supabase PKCE often drops next from redirectTo). Generic /auth/callback also treats type=recovery or preserved next=/auth/reset-password as recovery. Invalid/expired → /forgot-password?error=reset	Automated
Set new password	Shared policy + mismatch checks; updateUser then sign-out success	Automated
Recovery session on reset page	Authenticated recovery session stays on /auth/reset-password (not bounced to /map by proxy)	Automated
Protected routes redirect	Unauthenticated → /login?next=	Automated
Callback exchange	Session established; onboarding-aware redirect	Partial automated;

(confirm)		Manual recommended
Logout	Session cleared	Automated / Manual

Vehicle / profile tests

Case	Expected	Coverage
Incomplete vehicle blocked from map/publish/claim	Redirect or VEHICLE_PROFILE_REQUIRED	Automated
Vehicle fields validation	Invalid plate/year rejected	Automated
Profile update	Display name / vehicle save	Automated / Manual
CarImages fallback	Generic illustration when unavailable	Automated UI tests

Publishing tests

Case	Expected	Coverage
Publish Now	handoff_started_at set; expires_at $\approx$ start + 3 minutes	Automated helpers + Manual
Publish future (e.g. +5)	Unstarted; expires_at $\approx$ available_at	Automated helpers + Manual
One open spot per owner	Second open publish blocked	DB constraint; Manual
Cancel with structured reason	Terminal + reason stored; no credits	Automated + Manual

Discovery / map tests

Case	Expected	Coverage
Available spots appear	status = available and expires_at > now	Automated discovery merge tests
Claimed pin removed / tombstone	Hidden for seekers	Automated
Realtime upsert after unclaimed I'm	Pin stays; expires_at updates	Automated

leaving now		
Initial map camera — trusted GPS over cache/default	Current trusted fix preferred over stale cache or Tel Aviv fallback (<code>resolve-initial-map-camera</code>)	Automated
Initial map camera — no Herzliya/Sokolov boot pin	Map does not start on old hardcoded development coordinates	Automated
Initial map camera — late GPS before interaction	If GPS arrives shortly after mount, map recenters only while the user has not panned/dragged	Automated (<code>ParkingMapMapLibre.geolocation.test.tsx</code>)
Initial map camera — user interaction wins	After manual pan/drag, a later GPS fix does not unexpectedly recenter the map	Automated
Initial map camera — lat/lng order	MapLibre [<code>lng</code> , <code>lat</code>] order preserved; swapped coordinates rejected as out of bounds	Automated
Initial map camera — Tel Aviv fallback	Safe Tel Aviv fallback used only when no better in-bounds location is available	Automated
Share a Spot — empty coordinate strings	Empty <code>latitude</code> / <code>longitude</code> strings must not be treated as <code>0,0</code> ; publish blocked until a real location is set	Partial automated (UI <code>hasLocation</code> guard); Manual recommended
Share a Spot — late GPS vs manual pin	Fresh GPS must not overwrite an address selection or manual pin move	Automated (<code>PublishSpotForm.test.tsx</code>)
Map location providers	Android native and browser/PWA use different device location providers but share the same initial camera resolution logic	Automated helpers + Manual on real devices
Map without MapTiler key	Degraded	Manual
Primary map stack	MapLibre + MapTiler for seeker experience	Code inspection + Manual

Claim tests

Case	Expected	Coverage
Successful claim nearby	Spot claimed; active claim created	Automated action mocks + Manual
Already claimed / race loser	Business error (e.g. spot unavailable)	Automated mapping + Manual
Simultaneous claims	Exactly one active claim	DB FOR UPDATE + unique index; Manual strongly recommended
Expired / past available unstarted	SPOT_EXPIRED	Automated / migration contract
Insufficient credits	Rejected; no debit	Automated + Manual
Active claim exists for seeker	Rejected	Automated + Manual
Self-claim	Rejected	Automated / Manual
Too far (>1500 m)	Rejected	Automated claim-distance tests + Manual
Same-seeker reclaim after voluntary release	ALREADY_RELEASED_THIS_SPOT	Migration + action tests + Manual
Claim into live Now / early- start window	Uses remaining expires_at; timer not reset	Migration contract + Manual

Timing tests

Case	Expected	Coverage
Now listing	Live 3-minute claimable window	Automated helpers/UI + Manual
Future unclaimed at departure	Expires at available_at	Migration + Manual
Future claimed at departure	Auto-start 3-minute handoff	Migration + reconcile tests + Manual
I'm leaving now + existing claim	Starts claimed handoff from now(); 3- minute window	Automated start tests + Manual
I'm leaving now + no claim	Stays available; live window; claimable until expires_at	Migration 20260819140000 + Manual
Claim mid-window	Remaining time only	Countdown/render tests + Manual

+2 extension once	Caps at start + 5 minutes	Constants/extension tests + Manual
Double press I'm leaving now	Idempotent; no timer reset	Start action tests

Cancellation / release tests

Case	Expected	Coverage
Seeker release before start	Claim cancelled; spot may reopen for others; releaser cannot reclaim same spot; no credits	Manual + reclaim tests
Seeker release after start	Exchange ends; no automatic relist; no credits	Manual
Publisher cancel with reason	Ends listing/exchange; structured reason; no credits	UI CancelSpotButton tests + Manual

Completion / verification tests

Case	Expected	Coverage
Only publisher can complete	Non-owner rejected (NOT_OWNER)	Automated + Manual
Correct seeker plate suffix	Completes; seeker -1 / publisher +1 once	Automated + Manual
Incorrect suffix	Attempts remaining; no credit move	Automated + Manual
3 failures → cooldown	Locked ~2 minutes	Automated + Manual
Idempotent complete	Second successful call does not double-move credits	Automated complete tests
Complete while claimed but before live start and before available_at	HANDOFF_NOT_STARTED (auto-start only applies when due)	Manual
Complete while claimed and due (now >= available_at) but not yet started	Auto-start runs, then plate verification may proceed	Manual

Current product path: **publisher verifies seeker last-2 plate digits**. Spoken 5-digit codes are **not** the product UX.

Database tests

Case	Coverage
Partial unique indexes for active claims / ledger	Migration SQL + contract tests
Status CHECKs	Schema migrations
RLS "own row / participants"	Documented in migrations; full live RLS suite not claimed as exhaustive automated coverage
RPC authorization (auth.uid)	Migration bodies; Manual with two users
Transactions / locking	Implemented in SQL; Manual race demos recommended

Permissions / security tests

Case	Coverage
Cannot update another user's profile	RLS; Manual
Cannot complete another user's claim as non-owner	RPC; Automated mocks + Manual
Masked plate to counterpart	Counterpart vehicle RPC/UI; Automated + Manual
Anon cannot call privileged RPCs	Grants in migrations; Manual / SQL
get_handoff_code revoked from authenticated	Migration contract (20260823100000); Manual SQL
Live-location atomic rate limit (GPS + status)	Migration contract + algorithm regression tests; Edge Function contract tests
Broadcast only after DB accept	Edge Function contract tests
Auth redirect rejects open redirects	safe-redirect.test.ts
Baseline security headers	security-hardening.test.ts (reads next.config.ts)
Native/JS diagnostics do not log JWTs	Code review + logging key filters

History tests

Case	Coverage
First page load	Automated load-history tests
Load more cursor	Automated
Only own terminal handoffs	Manual + RPC design

Error handling tests

Case	Coverage
Business error toast mapping	Automated (error-map, claim button tests)
Route error boundary	Automated (RouteErrorScreen.test.tsx)
Stale chunk / failed dynamic import	One auto-reload then stop (stale-client-build.test.ts)
Unrelated exception does not auto-reload	Automated
Network failure	Manual

UI tests

Many component tests exist (sheets, countdown, publisher card, map carousel, nav, etc.).

They do **not** replace a real-device GPS demo.

Case	Expected	Coverage
Header credits display	Same profiles.credits; hidden on auth routes	Automated (HeaderCreditsBalance, header-credits-auth)
Completion success overlay / return to map	Overlay at app root; refresh vs replace; map-ready dismiss	Automated
Terminal ended overlay	Cancel / release / expiry copy; no credit change	Automated

Edge cases (checklist)

- Browser refresh mid-handoff (state restores from DB).
- Offline page via service worker (PWA).
- Claim vs early-start near-simultaneous.
- Extension near hard cap.
- Publisher verifies wrong digits then succeeds after cooldown.
- Seeker with 0 credits.
- Second device Realtime lag then refresh.
- Live location permission denied (claim still exists; sharing degraded).
- Web live location: seeker posts via Edge Function `handoff-seeker-location` ; publisher receives via private Broadcast + DB snapshot poll.
- Web foreground required for continuous GPS in browser tabs (see matrix below).
- Long-lived tab after a new Vercel deploy: one stale-chunk recovery reload, then the fatal page if still broken.
- Auth session refresh failure on resume: live location degrades; protected routes still require a session.
- Signup with an already-registered email (neutral Check your email; may receive no mail).

- Password-reset email must use `/auth/callback/recovery` (not rely on `next` surviving the Supabase PKCE redirect).
 - Expired password-reset link → request a new reset link from `/forgot-password`.
-

Manual end-to-end testing (two users)

Use two real accounts (A publisher, B seeker). Prefer physical devices with GPS.

Setup

- ☐ Both accounts have complete vehicles and ≥ 1 credit.
- ☐ MapTiler + Supabase env configured.
- ☐ Latest migrations applied on the Supabase project.

Core handoff (happy path)

- ☐ A publishes **Leaving in 2–5 minutes**.
- ☐ B sees pin; claims successfully.
- ☐ B opens navigation chooser.
- ☐ A presses **I'm leaving now** (or waits for auto-start).
- ☐ A sees B vehicle (masked plate); live location appears if GPS/permission allow.
- ☐ A extends once (optional).
- ☐ A enters **correct** last 2 digits → complete.
- ☐ Credits: B -1 , A $+1$; both History updated.

Negative / alternative

- ☐ Forgot password → reset email → `/auth/callback/recovery` → set new password → sign out → sign in (optional demo).
 - ☐ Second seeker C cannot claim A's already-claimed spot.
 - ☐ B with 0 credits cannot claim.
 - ☐ B too far cannot claim.
 - ☐ B releases before start → spot can return for others; B cannot reclaim same spot.
 - ☐ Wrong plate digits thrice → temporary lock (do this outside the presentation demo if possible).
 - ☐ Unclaimed future listing expires at departure without I'm leaving now.
 - ☐ Unclaimed **I'm leaving now** stays claimable for ~3 minutes.
-

Web live-location compatibility matrix (manual — real devices)

Roles: Seeker (driver) **sends** GPS; Publisher (spot owner) **receives** marker.

Foreground target: both users keep Switch It active in the browser tab.

Diagnostics: filter console for `[switch-it:handoff-live]` (seeker) and `[switch-it:handoff-live-receiver]` (publisher). Never log JWTs.

Case	Sender device/browser	Receiver device/browser	Publisher/seeker roles	GPS permission	Foreground result	
A	Android Chrome (website)	iPhone Safari (website)	B seeker → A publisher	Both allow precise location	Seeker: edge publish succeeded; Publisher: event received	SL or pu
B	iPhone Safari (website)	Android Chrome (website)	B seeker → A publisher	Both allow	Same as A, roles reversed	Si
C	iPhone Safari	iPhone Safari	B seeker → A publisher	Both allow	Same	Si
D	Android Chrome	Android Chrome	B seeker → A publisher	Both allow	Same	Si
A'	iPhone Safari	Android Chrome	A publisher → B seeker (receive-only on A)	Both allow	Publisher receive path only on A; B must still send when seeker	Pi or st se B pu

Known browser limitations (not bugs):

- Ordinary website tabs cannot guarantee GPS while user is in Waze/another app or after screen lock — native Android FGS remains the guaranteed background path.
- PWA installed vs browser tab: same web geolocation rules; installation is not required for foreground live location.

Automated coverage (Vitest): web runtime selects `navigator.geolocation` + Edge Function transport; native path unchanged; payload parser platform-independent; publisher snapshot poll while waiting; seeker watcher lifecycle; permission denied degradation.

Current automated testing summary

Metric	Value
Command	<code>npx vitest run (also npm run test:run)</code>

Framework	Vitest 4.x + Testing Library + jsdom
Test files	277
Tests	1744 (all passing on this documentation pass)
ESLint	<code>npx eslint . --pass (exit 0)</code>
Production build	<code>npm run build -- pass (Next.js 16.2.12)</code>
Playwright / Cypress project	Not currently implemented

Repository references

- `package.json`
- `src/**/*.test.ts(x)`
- `supabase/migrations/*.migration.test.ts`
- `src/actions/auth*.test.ts` , `src/lib/auth/auth-callback-handler.test.ts`
- `src/actions/claims*.test.ts` , `spots.start.test.ts`
- `src/lib/map/seeker-discovery-spots.test.ts`
- `src/lib/map/resolve-initial-map-camera.test.ts`
- `src/components/map/ParkingMapMapLibre.geolocation.test.tsx`
- `src/components/spots/PublishSpotForm.test.tsx`
- `src/lib/history/load-history.test.ts`
- `src/lib/feedback/error-map.ts`
- `src/lib/navigation/stale-client-build.test.ts`
- `src/components/shell/RouteErrorScreen.test.tsx`
- `src/lib/location/use-seeker-live-location-share.test.ts`
- `src/lib/realtime/use-realtime-invalidation.test.ts`
- `supabase/migrations/20260818210000_auto_start_handoff_at_departure.sql`
(`HANDOFF_NOT_STARTED`)